

# TECHNICAL DESIGN REPORT

## Simple Quadruped Robot Dog

Prepared for: Smart Methods Summer Training | Date: July 2026

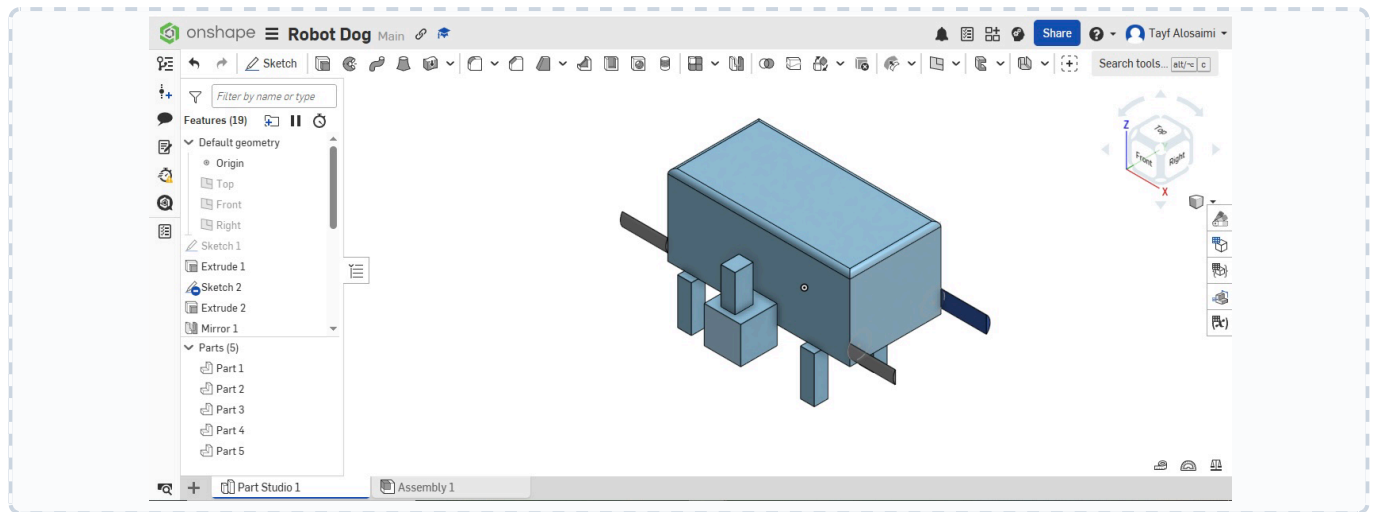
### 1. Introduction

This project presents the initial mechanical design of a simple quadruped robot dog. The model was created using Onshape as part of the summer training tasks. The main goal was to design a robot that can stand, keep its balance, and perform basic walking movements.

### 2. Mechanical Design

The robot has a simple and symmetrical design.

- The body is a rectangular chassis with smooth edges (Fillet).
- The robot has four legs placed evenly on both sides of the body.
- A simple head and tail were added to make the robot shape clearer.
- The design is kept simple because the goal of this task is to understand the basic mechanical principles of a quadruped robot.



### 3. Joints and Degrees of Freedom (DOF)

Each leg has 2 joints:

- Hip Joint
- Knee Joint

Since the robot has four legs:

$$2 \text{ joints} \times 4 \text{ legs} = 8 \text{ Degrees of Freedom (8 DOF)}$$

This number of joints is enough for basic walking while keeping the design simple.

## 4. Motor Selection

The selected motor is the MG996R Servo Motor. This motor was chosen because:

- It provides good torque.
- It is commonly used in robotics projects.
- It is suitable for moving the robot legs.

## 5. Torque Calculation

A simple torque calculation was performed for one joint.

### Assumptions:

- Robot mass = 2 kg
- Gravity = 9.8 m/s<sup>2</sup>
- Leg length = 0.05 m

### Total Force:

$$F = 2 \times 9.8 \approx 20 \text{ N}$$

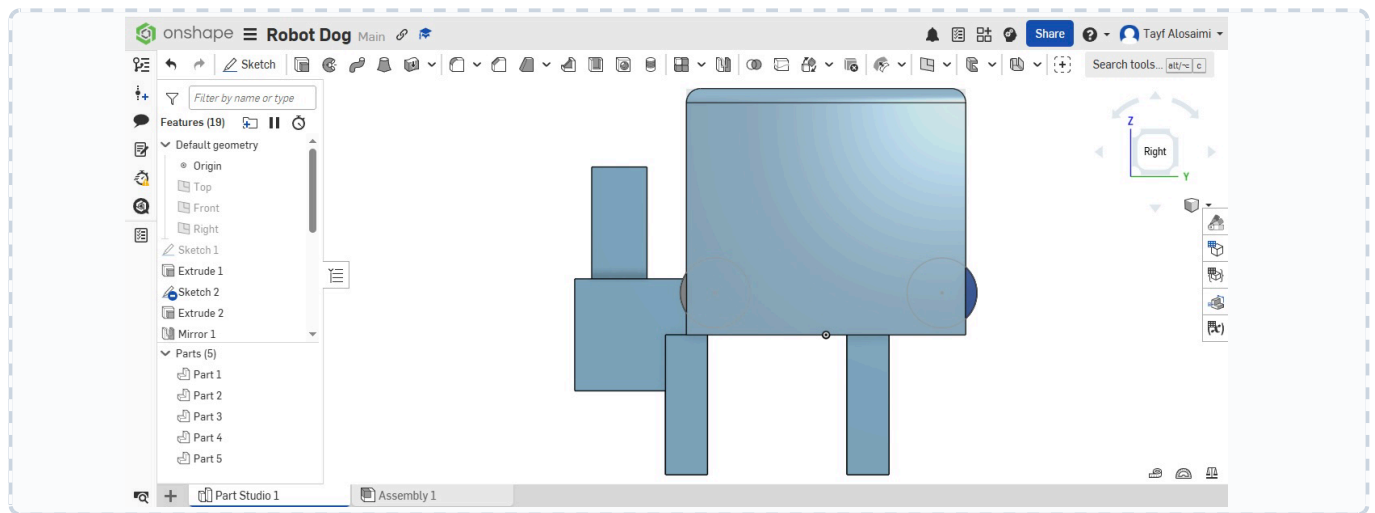
Assuming two legs support the robot while walking:

$$\text{Force on one leg} = 10 \text{ N}$$

### Torque:

$$\begin{aligned}\text{Torque} &= \text{Force} \times \text{Distance} \\ \text{Torque} &= 10 \times 0.05 \\ \text{Torque} &= 0.5 \text{ N}\cdot\text{m}\end{aligned}$$

Therefore, a servo motor with about 10 kg·cm torque is a suitable choice.



## 6. Stability and Center of Gravity

The robot has a symmetrical structure, which helps improve its balance. The battery and electronic components should be placed near the center of the body to keep the center of gravity balanced during movement.

## 7. Walking Method

The proposed walking method is the Trot Gait. In this method:

- The front-left leg moves with the rear-right leg.
- The front-right leg moves with the rear-left leg.

This walking style provides good balance and is commonly used for quadruped robots.

## 8. Expected Mechanical Challenges

Some possible challenges include:

- Keeping the robot balanced while walking.
- Choosing motors with enough torque.
- Preventing the legs from slipping.
- Reducing the total weight of the robot.

## 9. Conclusion

This project presents a simple mechanical design for a quadruped robot dog. The design includes the body, legs, joints, motor selection, torque calculation, stability, and walking method. Although it is a basic model, it demonstrates the main mechanical concepts needed before building or programming a real robot.